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## SARDAR PATEL UNIVERSITY

M. Sc (Information Technology) Semester-I Examination (NC) April 2023

PS01CINT54 : Relational Database Management System

19<sup>th</sup> April 2023, Wednesday

Time: 10.00 am to 1.00 pm

Total Marks: 70

Q-1 Select an appropriate option for the following statement:

[08]

- 1 The software used to store, manage, query, and retrieve data stored in a relational database known as \_\_\_\_\_.  
[a] DBMS [b] RDBMS [c] Table [d] Query
- 2 \_\_\_\_\_ level of the three-tier architecture also known as a logical level.  
[a] Internal [b] Conceptual [c] External [d] None of these
- 3 A \_\_\_\_\_ is an association among several entities.  
[a] Relationship [b] Association [c] Set [d] Combination
- 4 \_\_\_\_\_ entity is reliant on another entity.  
[a] Strong [b] Weak [c] Complete [d] Dependent
- 5 \_\_\_\_\_ represents relationships between tables.  
[a] Primary key [b] CHECK [c] Foreign key [d] None of given
- 6 \_\_\_\_\_ declares a variable to have the same data type as that of a previously defined variable or a column in the table.  
[a] Variable [b] % type [c] declare [d] None of given
- 7 \_\_\_\_\_ cursor attribute returns number of records in a cursor.  
[a] % Rowcount [b] % count [c] % recordcount [d] None of given
- 8 \_\_\_\_\_ returns always one value to a caller.  
[a] Function [b] Procedure [c] Trigger [d] a and b both

Q-2 Answer the following questions in brief. (Attempt Any SEVEN)

[14]

- 1 Explain the term Logical Independence & Physical independence.
- 2 Explain the terms: stored and derived attributes..
- 3 Compare DBMS with RDBMS.
- 4 List out at least FOUR notations used to draw E-R diagram with its brief purpose.
- 5 Differentiate clearly single value attribute and multi valued attribute.
- 6 Write a command to display the result of  $500 * 200$
- 7 Explain GROUP BY clause with an example.
- 8 Write a function named 'FSUM, which subtracts to given numbers say A and B.
- 9 State the differences between procedure and trigger.

**Q-3 Do as directed.**

[A] Write a note on users associated with database system with their roles.

[B] Discuss three level schema architecture of DBMS.

[06]

[06]

**OR**

[B] Briefly explain concept of data sharing, data integrity and data protection.

[06]

**Q-4 Do as directed.**

[A] List out Codd's rules and explain any two of them.

[B] Explain term Normalization. Discuss 1NF and 2NF with example.

[06]

[06]

**OR**

[B] What do you mean by Relationship? Discuss degree of relationship with example.

[06]

**Q-5 Do as directed.**

[A] Write SQL command for each of the following. [Any FOUR]

[06]

Assume that the table EMP having fields (ENO number(2,0), DNO number(1,0), ENAME varchar2(15), BASIC\_PAY Number(8,0).

i. Display unique records of a table EMP

ii. Create the table EMP having primary key ENO.

iii. List out DNO and DNO wise total BASIC\_PAY for all employees of EMP table.

iv. Remove record having DNO greater than 2 from a table EMP.

v. Retrieve all the employee information of table EMP in ascending order of ENAME.

[B] Explain briefly following statement with example of your choice:

[06]

a. FOR statement    b. IF Statement

**OR**

[B] Explain briefly PL/SQL block structure with purposes of each section.

[06]

**Q-6 Do as directed.**

[A] Write short note on Cursor, including its definition, types and attributes.

[06]

[B] Write short note on error handling in PL/SQL with example.

[06]

**OR**

[B] Assume that table named TELE has fields TNO Number(5), CNAME varchar2 (20), NCALLS number(5,0) and BILL\_AMT number (11,0).

[06]

Write PL/SQL code block to read TNO from the user of a table TELE, if TNO is found in table TELE then print the message 'RECORD FOUND', otherwise print the message that 'RECORD NOT FOUND'.

